Customer Churn Analysis Using Python

Project Overview

Analyzed customer churn data to understand customer retention, subscription behavior, revenue at risk, and factors associated with customer cancellations.

Tools & Technologies

- Python
- Pandas & NumPy
- Matplotlib & Seaborn
- SQLite
- Jupyter Notebook

Work Performed

- Extracted customer, subscription, and support data from SQLite.
- Cleaned datasets by handling missing values, duplicates, inconsistent categories, and data types.
- Merged multiple tables using customer IDs.
- Created new features such as churn flag, churn risk, and customer tenure.
- Calculated key business KPIs including churn rate, retention rate, ARPU, and revenue at risk.
- Performed churn analysis by subscription plan and customer demographics.
- Created visualizations using Matplotlib and Seaborn.
- Performed correlation analysis and generated pivot tables to identify customer behavior patterns.

Key Insights

- **Churn Rate:** 28.57%
- **Retention Rate:** 71.43%
- **Average Revenue Per User (ARPU):** 18.85
- **Average Customer Tenure:** 1,581 days
- **Highest Churn:** Basic plan (60%)
- **Lowest Churn:** Premium plan (14.29%)

- **Escalation Rate:** 19.05%
- **Escalation vs Churn Correlation:** 0.77

Action Items

• Check what happened in Karnataka – was there any increase in price, user complaints, tech issues, etc

• Did we increased subs price for Basic plan or recently made any changes – specially in sep month?

• Check what competitor is doing – bcoz one user has moved to competitor

• Focus on customers with 'High' & 'Med' churn risk, check their LTV (to make priority list), complaint request, reach out to

them through email, sms, calls and fix their issue.

Conclusion

This project provided practical experience in data cleaning, data integration, exploratory data analysis, KPI calculation, and data visualization. The analysis highlighted differences in churn across subscription plans and an association between customer escalations and churn.